# Einstein-defect bridges without a Kähler assumption

23 September 2026. Two conclusions are distinguished: control of the **original metric** uses the scalar-band variance estimate and its spectral factor; control of a **conformal Yamabe representative** needs neither. The metrics in both statements need not initially be Kähler. The all-metric absolute-energy selection theorem is in the companion energy-selection paper, not repeated here.

## 1. Common hypotheses and exact accounting

Let $(M, g)$ be a closed connected oriented smooth four-manifold. A specified Spin$^c$ class $a$ is assumed to have ordinary Seiberg–Witten solutions for every metric conformal to $g$. The class may be selected separately for each metric, but is fixed as its conformal weight varies. Write

$$p = (a_g^+)^2 > 0, \quad A = a^2, \quad T = 2\chi + 3\tau, \quad V = \mathrm{Vol}(g),$$

$$S_2 = \int s^2, \quad P = \int |\mathrm{Ric}^\circ|^2, \quad L = \int (s - \bar{s})^2, \quad R_2 = P + S_2/4,$$

$$D = \frac{1}{4\pi^2} \int \left( \frac{s^2}{24} + \frac{|W^+|^2}{2} \right) - \frac{p}{2}. \tag{1}$$

Weyl norms are Hilbert–Schmidt operator norms. Classical monopole inequalities give $D \geqslant 0$ and $B_p := S_2 - 32\pi^2 p \geqslant 0$. Chern–Weil then gives the exact identities

$$R_2 = 8\pi^2(2p - T) + 32\pi^2 D, \qquad P + B_p/4 = 8\pi^2(p - T) + 32\pi^2 D. \tag{2}$$

These identities and the scalar inequality are classical inputs, not new existence results. Normalized weighted derivations are given in normalized monopole derivation, Section 6.

## 2. The original-metric bridge

Assume $-S \leqslant s \leqslant -s_0 < 0$, and put

$$K = S/\sqrt{s_0}, \quad C_0 = (1 + 3S/(2s_0))^2, \quad C_{\mathrm{var}} = \frac{2}{3} + \frac{4C_0}{3} + \frac{2K^2}{\lambda_1(g)}. \tag{3}$$

Here $\lambda_1(g) > 0$ is the first positive scalar Laplace eigenvalue. The actual-form remainder proved in scalar-band remainder, equation (11), gives $L \leqslant 144\pi^2 C_{\mathrm{var}} D$.

Let $\kappa = -\bar{s}/4 > 0$ and $Q_\kappa = \int |\mathrm{Ric} + \kappa g|^2 = P + L/4$. **Theorem 1.** Without assuming any Ricci bound,

$$Q_\kappa \leqslant 8\pi^2(p - T) + (32\pi^2 + 36\pi^2 C_{\mathrm{var}})D. \tag{4}$$

In particular, define the nonnegative supply

$$\mathscr{P} = (p - A) + (A - T)_+, \quad U = 8\pi^2 \mathscr{P} + (32\pi^2 + 36\pi^2 C_{\mathrm{var}})D. \tag{5}$$

For the relative integral Einstein defect one has

$$\boxed{Q_\kappa \leqslant U, \qquad \mathscr{E}(g) := \inf_{\lambda \neq 0} \frac{\int |\mathrm{Ric} - \lambda g|^2}{\lambda^2 V} \leqslant \frac{4U}{4\kappa^2 V + U} \leqslant \frac{16U}{s_0^2 V}.} \tag{6}$$

**Proof.** Drop the nonnegative $B_p/4$ in (2), and add the scalar-variance bound divided by four. Also $p - A = \|a_g^-\|_2^2 \geqslant 0$, hence $p - T \leqslant \mathscr{P}$. Direct minimization in $1/\lambda$ gives $\mathscr{E} = 4Q_\kappa/(4\kappa^2 V + Q_\kappa)$; $\kappa \geqslant s_0/4$ completes the proof.

For a sequence in a fixed scalar band, sufficient conditions are $\mathscr{P}/V \to 0$, $D/V \to 0$, and $D/(\lambda_1 V) \to 0$. A uniform spectral gap suffices for the last condition but is not mandatory. It cannot simply be deleted. At $D = \mathscr{P} = 0$, the conclusion is negative Kähler–Einstein; $D = 0$ alone gives only cscK. Unequal-curvature hyperbolic products have $D = 0$ and $p - A > 0$.

## 3. Supplying an actual form and its joint analytic budget

Constant-rescale to $\bar{s} = -4$, and use the scalar-band parameters of that normalization. Then $r := \fint |\mathrm{Ric} + g|^2 \leqslant U/V$. Suppose $U/V \leqslant r_0$, $D/V \leqslant d_0$, and $(-a^2)_+/V \leqslant a_0$. Sections 6–7 of the scalar-band note supply a smooth actual weighted-monopole form $\omega$, for every prescribed $\eta > 0$, with

$$\|\omega\|_\infty \leqslant \frac{\sqrt{3}}{2} K, \quad B_\omega := \fint |\Delta_H \omega|^2 \leqslant C(S, s_0, 8 + 2r_0, a_0, d_0), \tag{7}$$

$$\begin{aligned} A_{\omega,1/2} &= \frac{1}{2}\left(\frac{|\omega|^2}{3} I - \omega \otimes \omega\right), \\ d_{1/2} &:= \fint \left(|\mathrm{Ric} + g|^2 + |W^+ - A_{\omega,1/2}|^2 + (|\omega|^2 - 2)^2\right) \\ &\leqslant \frac{4U}{V} + \frac{16\pi^2(10C_0 + 8)}{3}\frac{D}{V} + \eta. \end{aligned} \tag{8}$$

Consequently

$$(1 + B_\omega) d_{1/2} \leqslant C\left(\frac{U}{V} + \frac{D}{V} + \eta\right). \tag{9}$$

The constant in (9) depends only on the displayed budgets. To check the inputs, $\fint |\mathrm{Ric}|^2 \leqslant 8 + 2r_0$; the bounded-spinor Hessian estimate controls $\Delta_H \omega$ without derivatives of the conformal weight or $W^-$. This is an explicit invocation of the inherited Hessian lemma, not a consequence of finite algebra checks. If the chosen class additionally satisfies $a^2 \geqslant T$, Chern–Weil gives $(-a^2)_+/V \leqslant r_0/(8\pi^2)$, so no independent square-cost bound is needed in this special case.

Unlike a circular composition, Ricci control in (7)–(9) is an **output** of Theorem 1. However, 1/2 is the realizable Kähler–Einstein root in Round21; (9) is a calibrated-form and source-budget statement, not a forbidden-root contradiction or an exact metric repair.

## 4. A conformal bridge without scalar-band or spectral assumptions

Return to only the hypotheses of Section 1. Define

$$\theta(g) = \frac{p - T + 4D(g)}{p} \geqslant 0. \tag{10}$$

Nonnegativity follows from (2). **Theorem 2.** The conformal class has a negative constant-scalar-curvature Yamabe representative $\widehat{g}$ satisfying

$$\boxed{\mathscr{E}(\widehat{g}) \leqslant \frac{4\theta(g)}{1 + \theta(g)}.} \tag{11}$$

**Proof.** The Yamabe theorem supplies a minimizer. Its scalar curvature cannot be nonnegative because the same conformal Spin$^c$ structure has $p > 0$ and an actual solution. Thus its Yamabe constant $Y < 0$. Normalize $\widehat{g} = u^2 g$ by $\int u^4 = 1$. Then

$$Y = \int (6|du|^2 + su^2) \geqslant -\|s_-\|_2, \quad S_2(\widehat{g}) = Y^2 \leqslant S_2(g). \tag{12}$$

The quantities $p$ and $\int |W^+|^2$ are conformally invariant, so $D(\widehat{g}) \leqslant D(g)$. Since $\widehat{g}$ has constant scalar curvature, $\mathscr{E}(\widehat{g}) = 4P(\widehat{g})/R_2(\widehat{g})$. By (2) and $B_p(\widehat{g}) \geqslant 0$, this is at most

$$\frac{4[p - T + 4D(\widehat{g})]}{2p - T + 4D(\widehat{g})}.$$

The expression increases with $D$, its derivative having the sign of $p > 0$. Replacing $D(\widehat{g})$ by $D(g)$ proves (11).

Theorem 2 changes the metric conformally; it does not remove the spectral term from Theorem 1 on the original metric. Neither conclusion upgrades integral approximation to pointwise Ricci pinching or to an Einstein metric. Relative to Round 17, the direction is reversed: Ricci and scalar errors are derived here from actual period/topological data and LeBrun saturation instead of being supplied as hypotheses.

Sources: LeBrun, Propositions 4.1–4.2, the classical Yamabe theorem, the scalar-band remainder cited above, and the normalized monopole inequalities. The algebra checker checks finite algebra, including the companion energy-selection identities; it does not certify these PDE inputs.